DAY 12 | HANDS-ON ACTIVITIES

React UI Logic and API Integration: Problem Statements

Seven problems building a full stack Task Board. In Colab, use show_react() and the served app from the notebook.

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How to work: read the problem, write the steps in plain English first, then write the code. Test with the sample run, then try your own inputs, including wrong ones.

Problem 1: Login Banner

Level: Warm-up | **Concepts:** ternary, &&

Build a component with a loggedIn state and a toggle button. Show "Welcome back, Mei" or "Please log in", and show a red badge with unread count only when it is above 0.

Sample run / expected result

```
Welcome back, Mei  [3 unread]
... toggle ...
Please log in
```

Hints

- {unread > 0 && ...}

Problem 2: Product List

Level: Easy | **Concepts:** map, keys

Render an array of 6 products as cards, sorted by price. Products with stock 0 show "Out of stock" and a greyed style.

Sample run / expected result

```
Pen Rs 20
Mug Rs 250 (Out of stock)
...
```

Hints

- Sort a copy: [...products].sort(...)

Problem 3: Search and Filter

Level: Medium | **Concepts:** controlled input, derived list

Add a search box and a "Hide done" checkbox to a task list. Show "No tasks match" when the filtered list is empty.

Sample run / expected result

```
Search: "work" -> Finish workbook
No tasks match
```

Hints

- Derive visible tasks during render; no extra state.

Problem 4: Four States

Level: Medium | **Concepts:** useEffect, loading, error, empty

Create a fake API function that resolves after 1 second, randomly fails 30 percent of the time, and sometimes returns an empty list. Show a loading, error (with Retry), empty and data view.

Sample run / expected result

```
Loading...
Could not load tasks [Retry]
No tasks yet
3 tasks loaded
```

Hints

- Retry can increment a counter used in the useEffect dependency array.

Problem 5: Tasks API

Level: Medium | **Concepts:** FastAPI CRUD, CORS

Build GET, POST, PATCH and DELETE /api/tasks in FastAPI with Pydantic models and CORS for http://localhost:5173. Test with TestClient.

Sample run / expected result

```
POST /api/tasks {"title": "Demo"} -> 201
PATCH /api/tasks/1 {"done": true} -> 200
```

Hints

- CORSMiddleware(allow_origins=[...])

Problem 6: Connect React to FastAPI

Level: Challenging | **Concepts:** fetch GET and POST

Load tasks from the API on mount, add new tasks with POST, toggle done with PATCH and delete with DELETE. Disable the Add button while saving.

Sample run / expected result

```
Adding a task shows it immediately after the 201 response
```

Hints

- setTasks(prev => [...prev, created])

Problem 7: Optimistic Toggle

Level: Stretch | **Concepts:** optimistic UI, rollback

Make toggling done instant: update state first, then PATCH. If the request fails, revert the change and show an error message for 3 seconds.

Sample run / expected result

```
Toggle -> instant tick -> API fails -> tick removed + "Could not save"
```

Hints

- Keep the previous list in a variable before changing state.

Ganbatte (頑張って): do your best